=== CONFIGURACION GNB (1779215644) ===


```yaml
# This example configuration outlines how to configure the srsRAN Project gNB to create a single FDD cell
# transmitting in band 3, with 10 MHz bandwidth and 15 kHz sub-carrier-spacing.
# This specific example uses ZMQ in place of a USRP for the RF-frontend.
# This configuration is intended to be used with srsUE.

cu_cp:
  amf:
    addr: 10.53.1.2               # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.3            # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "20893"
            tai_slice_support_list:
              - sst: 1 # SST 1: eMBB (enhanced Mobile Broadband) as per 3GPP TS 23.501
  inactivity_timer: 7200            # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 72

ru_sdr:
  device_driver: zmq                # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
  srate: 11.52                      # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75                       # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75                       # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 368500                  # ARFCN of the downlink carrier (center frequency).
  band: 3                           # The NR band.
  channel_bandwidth_MHz: 10         # Bandwith in MHz. Number of PRBs will be automatically derived.
  common_scs: 15                    # Subcarrier spacing in kHz used for data.
  plmn: "20893"                     # PLMN broadcasted by the gNB (MCC 208, MNC 93).
  tac: 7                            # Tracking area code (needs to match the core configuration).
  pdcch:
    common:
      ss0_index: 0                  # Set search space zero index to match srsUE capabilities
      coreset0_index: 6             # Set CORESET Zero index to 6 for 10MHz bandwidth as per instruction
    dedicated:
      ss2_type: common              # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1           # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64                # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64                # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log            # Path of the log file.
  all_level: info                   # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false                 # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap   # Path where the MAC PCAP is stored.
  ngap_enable: false                # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true                 # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap        # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap  # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap  # Path where the CU-UP E2AP PCAP is stored.

metrics:
  layers:
```

```
    enable_rlc: true                # Enable RLC metrics
    enable_sched: true              # Enable DU scheduler metrics
  periodicity:
    du_report_period: 1000          # Set DU statistics report period to 1s
```

=== CONFIGURACION UE (1779215644) ===

```ini
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc  = 63BFA50EE6523365FF14C1F45F88737D
k    = 00112233445566778899aabbccddeeff
imsi = 208930123456780 # MCC 208, MNC 93

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER COMPOSE (1779215644) ===

```yaml
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
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#

services:
  mongodb:
    container_name: open5gs_mongodb
    image: mongo:4.4
    networks:
      ran:
        ipv4_address: 10.53.1.100 # Assign an IP for MongoDB within the ran network
    healthcheck:
      test: echo 'db.runCommand("ping").ok' | mongo mongodb:27017/test --quiet
      interval: 3s
      timeout: 1s
      retries: 60

  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    env_file:
      - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
    privileged: true
    ports:
      - "9898:9999/tcp" # WebUI port
      - "38412:38412/sctp" # AMF NGAP port
      - "2152:2152/udp" # UPF GTP-U port
    volumes:
      - ./open5gs-5gc.yml:/open5gs-5gc.yml # Mount the generated core config
    command: 5gc -c /open5gs-5gc.yml
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 10.53.1.2 38412" ] # Check AMF NGAP port
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
    depends_on:
      - mongodb # Ensure MongoDB is up before 5GC

  gnb:
```

```yaml
    container_name: srsran_gnb
    image: srsran/gnb
    build:
      context: .
      dockerfile: docker/Dockerfile
      args:
        OS_VERSION: "24.04"
    privileged: true
    cap_add:
      - SYS_NICE
      - CAP_SYS_PTRACE
    volumes:
      # Access USB to use some SDRs
      - /dev/bus/usb/:/dev/bus/usb/
      # Sharing images between the host and the pod.
      # It's also possible to download the images inside the pod
      - /usr/share/uhd/images:/usr/share/uhd/images
      # Save logs and more into gnb-storage
      - gnb-storage:/tmp
    # It creates a file/folder into /config_name inside the container
    # Its content would be the value of the file used to create the config
    configs:
      - gnb_config.yml
      - gnb_compose_config.yml
    # Customize your desired network mode.
    # current netowrk configuration creastes a private netwoek with both containers attached
    # An alterantive would be `network: host"`. That would expose your host network into the container. It's the easie
    networks:
      ran:
        ipv4_address: ${GNB_IP:-10.53.1.3}
      metrics:
        ipv4_address: 172.19.1.3
    # Start GNB container after 5gc is up and running
    depends_on:
      5gc:
        condition: service_healthy
    # Command to run into the final container
    command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: ./gnb.yaml # Path to your desired config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: ${GNB_IP:-10.53.1.3}
      metrics:
        autostart_stdout_metrics: true
        enable_json: true
      remote_control:
        bind_addr: 0.0.0.0
        enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
```

        - subnet: 172.19.1.0/24
```

```yaml
# open5gs-5gc.yml (Core Configuration)
db_uri: mongodb://mongodb/open5gs

logger:

global:
  parameter:
    # Disable 4G core functions as this is a 5G setup
    no_mme: true
    no_sgwc: true
    no_sgwu: true
    no_pcrf: true
    no_hss: true

amf:
  sbi:
    server:
      - address: 127.0.0.5 # Internal IP for SBI within the container
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777 # Internal IP for SCP within the container
  ngap:
    server:
      - address: 10.53.1.2 # External IP for gNB connection
  metrics:
    server:
      - address: 127.0.0.5
        port: 9090
  access_control:
    - plmn_id:
        mcc: 208
        mnc: 93
  guami:
    - plmn_id:
        mcc: 208
        mnc: 93
      amf_id:
        region: 2
        set: 1
  tai:
    - plmn_id:
        mcc: 208
        mnc: 93
      tac: 1
  plmn_support:
    - plmn_id:
        mcc: 208
        mnc: 93
      s_nssai:
        - sst: 1 # SST 1: eMBB (enhanced Mobile Broadband) as per 3GPP TS 23.501
  security:
      integrity_order : [ NIA2, NIA1, NIA0 ]
      ciphering_order : [ NEA0, NEA1, NEA2 ]
  network_name:
      full: Open5GS
  amf_name: open5gs-amf0
  time:
    t3512:
      value: 540

smf:
  sbi:
    server:

```yaml
      - address: smf.5gc.mnc093.mcc208.3gppnetwork.org # FQDN for NRF registration
    client:
      scp:
        - uri: http://127.0.0.200:7777
  pfcp:
    server:
      - address: 127.0.0.4 # Internal IP for PFCP within the container
    client:
      upf:
        - address: 127.0.0.7 # Internal IP for UPF within the container
  gtpu:
    server:
      - address: 127.0.0.4 # Internal IP for GTP-U within the container
  metrics:
    server:
      - address: 127.0.0.4
        port: 9090
  session:
    - subnet: 10.47.0.0/16
      gateway: 10.47.0.1
    - subnet: 2001:db8:face::/48
      gateway: 2001:db8:face::1
  dns:
    - 8.8.8.8
    - 8.8.4.4
    - 2001:4860:4860::8888
    - 2001:4860:4860::8844
  mtu: 1400

upf:
  pfcp:
    server:
      - address: 10.53.1.2 # External IP for PFCP from SMF (if SMF is external) or internal if SMF is in same containe
  gtpu:
    server:
      - address: 10.53.1.2 # External IP for GTP-U from gNB
  session:
    - subnet: 10.47.0.0/16
      gateway: 10.47.0.1
      dev: ogstun
    - subnet: 2001:db8:face::/48
      gateway: 2001:db8:face::1
      dev: ogstun
  metrics:
    server:
      - address: 127.0.0.7
        port: 9090

nrf:
  serving:
    - plmn_id:
        mcc: 208
        mnc: 93
  sbi:
    server:
      - address: nrf.5gc.mnc093.mcc208.3gppnetwork.org # FQDN for NRF registration

scp:
  sbi:
    server:
      - address: 127.0.0.200 # Internal IP for SCP within the container
        port: 7777
    client:
      nrf:
        - uri: http://nrf.5gc.mnc093.mcc208.3gppnetwork.org # FQDN for NRF

ausf:
  sbi:
```

```
    server:
      - address: ausf.5gc.mnc093.mcc208.3gppnetwork.org # FQDN for NRF registration
    client:
      scp:
        - uri: http://127.0.0.200:7777

udm:
  hnet:
    - id: 1
      scheme: 1
      key: @build_configs_dir@/open5gs/hnet/curve25519-1.key
    - id: 2
      scheme: 2
      key: @build_configs_dir@/open5gs/hnet/secp256r1-2.key
  sbi:
    server:
      - address: udm.5gc.mnc093.mcc208.3gppnetwork.org # FQDN for NRF registration
    client:
      scp:
        - uri: http://127.0.0.200:7777

pcf:
  sbi:
    server:
      - address: 127.0.0.13 # Internal IP for PCF within the container
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
  metrics:
    server:
      - address: 127.0.0.13
        port: 9090
  policy:
    - supi_range:
        - 208930000000001-208930999999999 # Example SUPI range for MCC 208, MNC 93
      slice:
        - sst: 1  # SST 1: eMBB (enhanced Mobile Broadband) as per 3GPP TS 23.501
          default_indicator: true
          session:
            - name: internet
              type: 3
              ambr:
                downlink:
                  value: 1
                  unit: 3
                uplink:
                  value: 1
                  unit: 3
              qos:
                index: 9
                arp:
                  priority_level: 8
                  pre_emption_vulnerability: 1
                  pre_emption_capability: 1
            - name: ims
              type: 3
              ambr:
                downlink:
                  value: 1
                  unit: 3
                uplink:
                  value: 1
                  unit: 3
              qos:
                index: 5
                arp:
                  priority_level: 1
```

```
                pre_emption_vulnerability: 1
                pre_emption_capability: 1
            pcc_rule:
              - qos:
                  index: 1
                  arp:
                    priority_level: 1
                    pre_emption_vulnerability: 1
                    pre_emption_capability: 1
                  mbr:
                    downlink:
                      value: 82
                      unit: 1
                    uplink:
                      value: 82
                      unit: 1
                  gbr:
                    downlink:
                      value: 82
                      unit: 1
                    uplink:
                      value: 82
                      unit: 1
              - qos:
                  index: 2
                  arp:
                    priority_level: 4
                    pre_emption_vulnerability: 2
                    pre_emption_capability: 2
                  mbr:
                    downlink:
                      value: 802
                      unit: 1
                    uplink:
                      value: 802
                      unit: 1
                  gbr:
                    downlink:
                      value: 802
                      unit: 1
                    uplink:
                      value: 802
                      unit: 1
nssf:
  serving:
    - plmn_id:
        mcc: 208
        mnc: 93
  sbi:
    server:
      - address: nssf.5gc.mnc093.mcc208.3gppnetwork.org # FQDN for NRF registration
    client:
      scp:
        - uri: http://127.0.0.200:7777
      nsi:
        - uri: http://127.0.0.10:7777 # Internal IP for NRF within the container
          s_nssai:
            sst: 1 # SST 1: eMBB (enhanced Mobile Broadband) as per 3GPP TS 23.501
bsf:
  sbi:
    server:
      - address: 127.0.0.15 # Internal IP for BSF within the container
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777

udr:
```

```yaml
  sbi:
    server:
      - address: 127.0.0.20 # Internal IP for UDR within the container
        port: 7777
    client:
      scp:
        - uri: http://127.0.0.200:7777
```